Analyzing the 2024 U.S. Presidential Elections: BERT-based Sentiment Classification of Spoken Content in Video Data

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ABSTRACT

While existing implementations of sentiment analysis in election-related research have continuously relied on text-based social media platforms to analyze public opinions, the use of video data remains underexplored. This study explores extending sentiment analysis to the spoken content of video data concerning political discourse related to the 2024 U.S. Presidential Elections. The research fine-tunes pre-trained language models BERT and its lightweight variant DistilBERT, for sentiment analysis task using preprocessed video transcriptions. The models are further evaluated in terms of accuracy and training time. Sentiment classification results are then validated by contrasting the average sentiment proportions of each candidate against the actual election outcomes in three critical swing states: *Arizona, Michigan, and Pennsylvania*. The experimental results indicate that while sentiment analysis of transcribed videos captures nuances of public opinions, it does not reliably match the actual election results. Moreover, the comparative evaluation of BERT and DistilBERT reveals that although BERT models have always achieved better accuracy, DistilBERT is observed to train twice as fast as BERT while preserving minimal difference in performance. This difference in training time and performance metric highlights the trade-off between model complexity and computational efficiency.

CCS Concepts

• **Computing methodologies → Artificial intelligence → Natural language processing.**

Keywords

Sentiment Analysis; U.S. Presidential Elections 2024; BERT; DistilBERT; Natural Language Processing.

ACM Reference Format:

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1. INTRODUCTION

In recent years, sentiment analysis has become a vital instrument for understanding public perception. This type of analysis involves the process of classifying the emotional tone of a given text data using machine learning techniques. As social media and online platforms increasingly influence how people communicate to discuss political issues, accurately analyzing social media sentiments may play a pivotal role in understanding what the general public perceives about the 2024 U.S. Presidential Elections.

Using sentiment analysis in election-related research has shown promising outcomes. Existing studies in this regard have presented that analyzing social media data could help forecast election results [1] [2]. The studies applied various methods from natural language processing (NLP) to machine learning techniques to analyze available text data, gaining insights into how voters feel about specific candidates or parties.

Online video platforms have redefined political discourse by presenting contextually rich content about the state of major political events, such as the 2024 U.S. Presidential Elections. However, analyzing sentiments in this format poses a challenge due to the unstructured nature of spoken content in video data. Extending sentiment analysis to video data is crucial to better understand the general opinion of the public. Current applications of sentiment analysis in election-related research have relied on text-based platforms like Twitter to gauge public opinion. But the analysis of video content, particularly transcribed speeches, interviews, and political analysis that reflects the general opinion of the public, remains underexplored. This poses a significant gap, considering that video data is rich in context, which is vital for understanding public sentiments during elections.

A new possibility to extend sentiment analysis to the spoken content of video data may be implemented with recent improvements in speech-to-text transcription technologies like the OpenAI Whisper [3] [4]. Sentiments from these transcriptions can be analyzed using pre-trained language model Bidirectional Encoder Representations from Transformers (BERT). This model has continuously demonstrated proficiency in text classification tasks when it is fine-tuned on labeled data, given its ability to understand word context in a bidirectional manner [5] [6]. Targeted sentiment analysis based on specific topics or entities mentioned

within a transcription corpus could be implemented with the use of topic modeling and coreference resolution-based approaches in the preprocessing phase. Neural Coreference Resolution has been used by other studies to identify appropriate references of entities in a text [7] [8].

Given the lack of election-related research that considers the analysis of spoken content in video data, this study aims to explore the use of the BERT language model for sentiment analysis on videos about the 2024 U.S. Presidential Elections. Our study seeks to analyze the sentiments expressed in these videos to understand how candidates are perceived across three different swing states. The nature of the dataset utilized for this research consists of transcription data from YouTube videos. The research then analyzes sentiments towards presidential candidates Donald Trump and Kamala Harris across three historically significant swing states: *Arizona, Michigan,* and *Pennsylvania* [9]. Models BERT and DistilBERT were fine-tuned for sentiment analysis task, and their performance was evaluated. The sentiment analysis result made by the best-performing model was later validated against the actual election outcome to assess the difference between our model's analysis and real-world outcomes.

2. RELATED WORKS

In order to analyze the sentiments from multimedia types of data, existing methods include the conversion of audio and video data to a suitable text format for text-level analysis. Several studies have explored the use of AI tools and automatic transcription services to generate a corpus. Research studies have utilized OpenAI Whisper, a Transformer-based automatic speech recognition system that can be integrated and modified for robust handling of various speech recognition tasks [3] [4]. Vasconcellos et al. [3] justified Whisper's capability in producing high-quality transcriptions given that it is trained on one-third of non-English data points. This allows Whisper to adapt independently to the speakers' accent, regionalism, and voice speed [3].

Sentiment Analysis involves the use of various techniques to assess the emotional tone of textual data. Included in this are lexicon-based methods, statistical methods, word embeddings, context-dependent word embeddings, and specialized models. Each method offers distinct strengths in analyzing sentiments from a given text dataset. Nugroho [10] used lexicon-based VADER to analyze tweets from the 2020 US presidential election, providing predictions that closely aligned with the actual results. While this offers a quick interpretable result based on a predefined dictionary, other studies have explored the use of statistical methods that scale text based on frequency. The study by Subowo et al. [11] utilized TF-IDF combined with the Naive Bayes classifier to analyze tweets for the 2024 Presidential Election in Indonesia, though it faced challenges such as language and cultural biases. A more advanced approach such as word embeddings has also emerged to offer a deeper semantic understanding of text data. NLP techniques such as Word2Vec were able to capture relationships between words. Studies utilized Word2Vec for sentiment classification in Bahasa Indonesia and Indonesian presidential election tweets, respectively [12] [13]. Both studies achieved high accuracy and precision compared to baseline and existing models, illustrating the effectiveness of word embeddings in nuanced sentiment analysis. Advancements in word embeddings have led to the development of context-dependent word embeddings, which have enhanced the interpretation of nuanced meanings. A study by Yang et al. [14] employed ELMo with RNNs to capture context-sensitive meanings, outperforming other models and highlighting the importance of understanding word context. Similarly, BERT and LSTM were used for analyzing tweets related to the 2020 US presidential election, showing BERT's effectiveness in providing accurate predictions due to its bidirectional context, while LSTM had limitations with sentiment neutrality [15]. Specialized models like ElecBERT are developed by further fine-tuning BERT on domain-specific data, such as election-related content. ElecBERT outperformed other models in analyzing tweets linked to the election by successfully capturing the complicated language utilized in such contexts, as demonstrated by Khan et al. [16].

Existing research in the context of text analysis has explored various methods to extract entity-specific text and filter relevant documents based on specific subjects or themes. Included in these methods are Topic Modeling and Neural Coreference Resolution. Some studies applied topic modeling with sentiment analysis on user reviews to identify key factors associated with positive and negative reviews [17] [18]. This allows for a more granular approach that clusters documents for topic-specific sentiment analysis. However, this method is not adaptable for entity-specific extraction of text from an unstructured text corpus, as it focuses on identifying general topics rather than filtering contextually relevant sentences that reference specific subjects. Existing studies address this limitation with coreference resolution. Xue et al. [7] improved document-level extraction using the NeuralCoref spaCy library to identify mentions of entities across multiple sentences. Parrolivelli et al. [8] applied coreference resolution to process reference mentions for genealogical relationship extraction task from an unstructured document. Overall, this approach can be used as a preprocessing method to identify text references of target subjects, to filter only relevant sentences about these entities.

A groundbreaking study entitled "Attention Is All You Need" authored by Vaswani et al. [19] was published to address the limitations of traditional sequence-to-sequence models with the Transformers architecture. This led to the development of the BERT model. Devlin et al. [6] introduced Bidirectional Encoder Representations from Transformers (BERT), a pretrained NLP model that captures the deeper meaning of language through bidirectional understanding. BERT has been widely explored in different research involving text classification tasks. Yu et al. [20] used BERT to classify policy-related text by fine-tuning it on 10,000 policy-labeled texts. BERT outperformed CNN and LSTM models, achieving a 93.25% F1 score [20]. BERT-based models are further explored in studies [21] [22] [23] demonstrating higher accuracy in text classification compared to traditional machine learning classifiers and sequence-to-sequence models. The studies justify the utilization of the BERT model given its reusable nature through transfer learning and strong performance in text classification. Continuous research could delve into how BERT performs against smaller models like DistilBERT and larger models like RoBERTa to further assess accuracy, the tradeoffs in model size, and computational efficiency in tasks related to the classification of spoken text data.

Sentiment analysis is continuously being utilized in election-related research to gauge public opinion about candidates and assess its possible correlation with election outcomes. Researchers analyzed social media platforms to provide insights about voters' sentiments towards a specific party or candidate [24] [25] [26]. A key finding highlights the importance of using regional context in political sentiment analysis, revealing that support for movements like QAnon is not confined to traditionally Republican states but is also present in Democratic-leaning areas such as California [26]. This finding justifies the need for state-level analysis to capture the nuanced and geographically dispersed nature of political sentiment.

3. METHODOLOGY

This research focuses on analyzing the sentiments from video transcriptions related to the 2024 U.S. Presidential Election. The following subsections present the process used for data collection, data preprocessing, and fine-tuning of the model. The sentiment analysis result made by the best performing model is then validated against the actual election outcomes.

3.1 Data Collection

YouTube videos related to 2024 U.S. presidential election discussing Donald Trump and Kamala Harris are collected and identified for a given state if their title or description contains either the full state name (e.g., "Arizona", "Michigan", "Pennsylvania"), its postal abbreviation (e.g., "AZ", "MI", "PA"), or a major city within that state. The target is to obtain between 5000 to 7000 candidate-related sentences, a range based on existing research for text classification [22] [23]. Videos are iteratively collected until the minimum threshold of candidate-related sentences is reached. Collected videos are organized by state and catalogued with metadata including video ID, title, duration, and URL.

3.2 Video-to-Text: Audio Extraction and Transcription

Audio extraction is performed using the FFmpeg Python library where each video file is converted into an MP3 audio track. OpenAI Whisper is utilized to transcribe the audio files into transcripts, usable for text analysis.

Figure 1. Video-to-Text Processing Pipeline

3.3 Initial Text Preprocessing

The transcript documents are split into sentences using NLTK's sentence tokenizer. Each sequential question-answer pairs found are merged into a single entry (e.g., "Do you like Trump?" + "Yes, I do." → "Do you like Trump: Yes, I do.") to preserve conversational context. The sentences are tagged with their state label for categorization.

3.4 Filtering Candidate-Related Sentences

Sentences are filtered using Neural Coreference Resolution, providing candidate-related sentences with direct mentions or references (e.g., pronouns) of "Donald Trump" or "Kamala Harris". Each filtered sentence is tagged with the corresponding entity candidate, which is later saved in a CSV file. Initially, the study utilized the topic modeling approach BERTopic to cluster sentences into subjects Trump and Harris. Given the technical limitations of the said approach in filtering sentences with direct entity mentions, the use of the NeuralCoref extension from the spaCy Python library is explored by the study.

3.5 Data Cleaning

After filtering, the dataset is refined to ensure quality. Specifically, the process includes:

- Spelling correction through manual review to fix transcription errors.
- Text normalization to standardize punctuation, capitalization, and special characters.
- Incoherent sentence removal to discard off-topic or nonsensical entries.

3.6 Manual Sentiment Annotation

Each sentence is labeled with a sentiment score of 1 (positive), 0 (neutral), or -1 (negative). Three expert annotators with a bachelor's degree in psychology are consulted to independently assign labels following a voting scheme adopted from [27]. Sentences with the majority votes determine the final sentiment, while sentences with tied votes or complete disagreement are discarded.

Table 1. Voting-based Annotation Assisted by Experts

Labeler 1	Labeler 2	Labeler 3	Final Label	Notes
1	1	0	Positive (1)	Majority
-1	-1	-1	Negative (-1)	Majority
1	0	0	Neutral (0)	Majority
1	-1	0	Tie	Discarded

Inter-rater reliability for the sentiment labels is assessed using Fleiss' Kappa, implemented via the *statsmodels* library. It yields $\kappa = 0.56$, implying a moderate level of agreement among annotators.

$$Kappa(k) = \frac{\bar{P} - \bar{P}_e}{1 - \bar{P}_e} \quad (1)$$

Table 2. Sample of the Final Annotated Dataset

Sentence	Candidate	State	Annotated Sentiment
We've got internal people saying Donald Trump is very likely to win Arizona.	Donald Trump	Arizona	1
Kamala Harris has been a prosecutor, attorney general, a senator, and now vice president.	Kamala Harris	Arizona	0
I think that Donald Trump would be a total disaster for our nation, and I cannot vote for him.	Donald Trump	Michigan	-1

The final preprocessed and annotated dataset produced a total of 5,745 sentences. The annotated dataset is partitioned into a 70-10-20 split: Training (70 %), Validation (10 %), and Test (20 %). This allocation is used in a similar BERT text classification study [32]. The following split ensures a sufficient validation set to inform hyperparameter tuning without compromising evaluation on unseen data [28].

3.7 Model Framework

The base model "bert-base-uncased" from HuggingFace is used with BertForSequenceClassification, which adds a classification head on top of the pre-trained BERT model. Each sentence is tokenized using BertTokenizer and passed into the model along with its corresponding sentiment label.

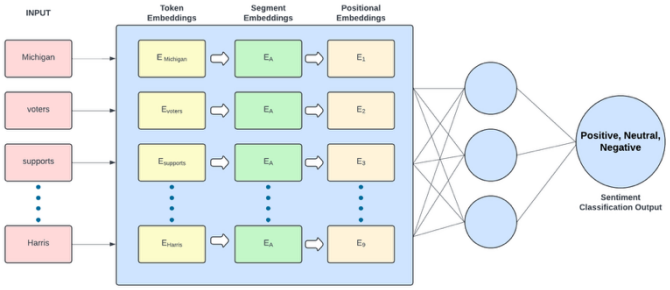

Figure 2. BERT Architecture for Sentiment Classification

To mitigate class imbalance, a weighted cross-entropy loss function is employed, with weights inversely proportional to class frequencies as in [29].

3.8 Fine-Tuning and Hyperparameter Optimization

The pre-trained BERT model, along with the added classification layer, is fine-tuned for sentiment classification task. During optimization, multiple hyperparameter configurations are experimented with.

Table 3. Hyperparameter Configurations

Learning Rate	Epochs	Batch Size
2e-5	2	16
3e-5	3	32
4e-5	5	32
2e-5	3	16

Each configuration is evaluated on the validation set for training loss, validation loss, and accuracy. The base learning rate of 2e-5 is experimented initially to minimize catastrophic forgetting [5].

3.9 Performance Assessment and Comparative Analysis

The overall model performance on the test set was evaluated using metrics: *Accuracy*, *Precision*, *Recall*, and *F1-score*. *Accuracy* measures the correctness of sentiment predictions. *Precision* quantifies the model's capability to avoid false positives. *Recall* measures the coverage of the model in capturing true positives. *F1-score* is the harmonic mean of precision and recall.

To benchmark performance, the fine-tuned BERT model is compared against DistilBERT, a lighter variant of the model. Contrary to previous studies that compare BERT against traditional machine learning methods, the research builds on further exploration of the base BERT against its modern versions like DistilBERT as seen in studies [22] [23]. The performance variations between the two models are evaluated in terms of the model's accuracy and training time. This head-to-head analysis examines the potential trade-off between model performance and training efficiency.

3.10 Comparison of Sentiment Predictions with Actual Election Results

To validate real-world relevance, sentiment predictions by our model are compared against the actual vote shares reported by CNN News [30] in the swing states included in this study.

First, for each candidate, the average sentiment is computed using (2) to have a value representation reflecting the general sentiment.

$$\bar{s} = \frac{\sum sentiment\ scores}{n} \tag{2}$$

Min–Max Normalization is then applied to rescale and normalize the average sentiment values using (3).

$$x' = \frac{x - \min(x)}{\max(x) - \min(x)} \tag{3}$$

The equation is then simplified to (4), since sentiment scores lie in the fixed bounds [-1, 1].

$$\bar{s}_{norm} = \frac{\bar{s} - (-1)}{1 - (-1)} = \frac{\bar{s} + 1}{2} \tag{4}$$

Next, the candidate's normalized sentiment score proportion is calculated using (5), wherein the normalized average sentiment score of the candidate is divided by the total pool of normalized average sentiments in a state.

$$p_i = \frac{\bar{s}_{norm,i}}{\sum_{n=1}^{2} \bar{s}_{norm,n}} \times 100 \tag{5}$$

This calculates the overall proportion of sentiment scores shared between candidates in a corresponding state, allowing comparison to actual vote shares.

$$Softmax(x_i) = \frac{e^{x_i}}{\sum_j e^{x_j}} \tag{6}$$

In addition, the Softmax equation is also explored as a normalization approach to consider how different scaling techniques and interpretations affect the sentiment result (6).

4. RESULTS AND DISCUSSION

4.1 Data Exploration

The study gathered a total of 915 videos from three different states. Specifically, 327 videos for Arizona, 320 for Michigan, and 268 for Pennsylvania. The total number of the final training dataset extracted from the videos consists of 5,745 relevant sentences, where 3,050 sentences referenced the subject Donald Trump, and 2,695 for Kamala Harris. These relevant sentences are annotated. The distribution of samples per sentiment class indicates that 39.3% of the final training dataset is negative, 28% is neutral, and 32.7% is labeled as positive. A weighted cross-entropy loss function was implemented in the classification layer during fine-tuning to account for the slight class imbalance in our final training dataset. Figure 3 illustrates the raw counts of sentences per sentiment class.

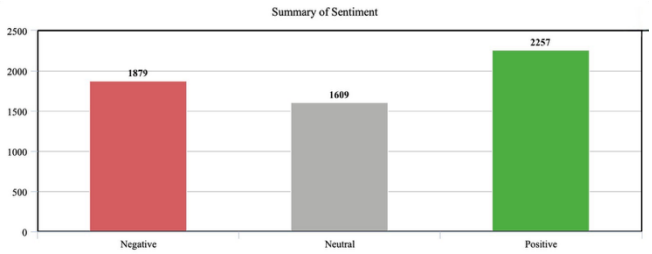

Figure 3. Summary of Sentiment Labels.

4.2 Model Validation

The validation results, as illustrated in Figures 4 and 5, suggest a constant decrease in training loss for all models. While this indicates that the models are learning, there is also a clear sign of overfitting shown after the second epoch. Validation of the BERT model suggests that overfitting is quickly evident in parameters with higher learning rates and longer training durations. After the second epoch, the validation loss tends to increase, suggesting that the model is failing to generalize on unseen data. The validation accuracy also began to plateau, which possibly indicates that the model stopped improving after the second epoch. For the DistilBERT model, a similar trend is developing wherein the model is learning as indicated by the decreasing training loss, but overfits drastically in higher learning rates and longer training durations. Validation accuracy also tends to plateau. For both models experimented with varying ranges of hyperparameters, lower learning rates with moderate epochs and small batch sizes tend to perform more effectively than higher learning rates.

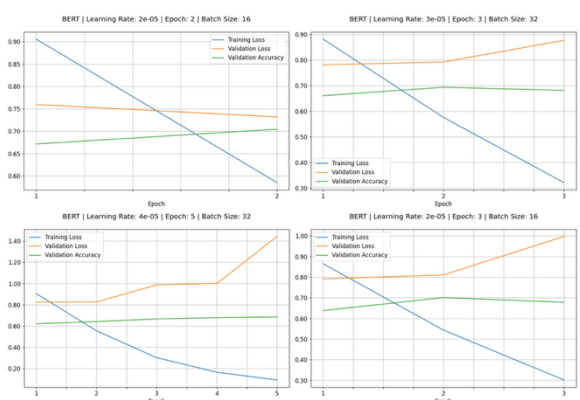

Figure 4. BERT Validation

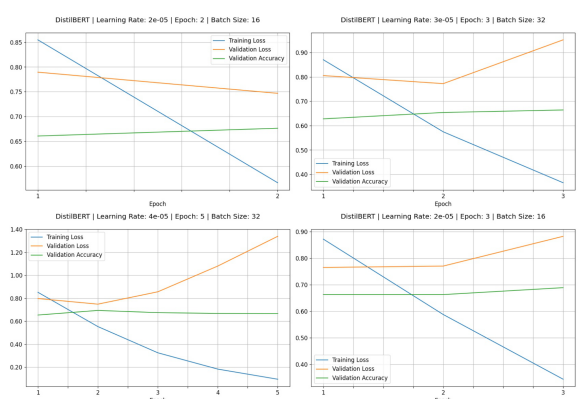

Figure 5. DistilBERT Validation

4.3 Model Evaluation and Comparison

The results of the evaluation for all trained models suggest minimal differences in performance between various hyperparameters. The main factor influencing the performance appears to be the size of the epochs and the learning rate. The combination of a higher learning rate and longer training duration resulted in a decrease in model performance. This implies that rapid learning by the model, even with longer training duration, does not equate to improved quality of the model on unseen data.

BERT slightly outperformed DistilBERT, achieving a ceiling accuracy of 70.84% compared to DistilBERT's 69.62%. The DistilBERT model achieved its highest accuracy using the same 2E-05 learning rate, but with more training runs at 3 epochs. This is possibly linked to its smaller parameter size, requiring more epochs to fully learn complex patterns than BERT's large capacity.

Table 4. Summary of Test Results

	Hyperparameter	Accuracy	Precision	F1-score	Recall	Training Time
BERT	LR: 2e-05 | E: 2 | BS: 16	0.7084	0.7009	0.7046	0.7084	39 min 1 sec
	LR: 3e-05 | E: 3 | BS: 32	0.7067	0.7227	0.7146	0.7067	1 hr 11 min 36 sec
	LR: 4e-05 | E: 5 | BS: 32	0.6927	0.6805	0.6866	0.6927	1 hr 59 min 20 sec
	LR: 2e-05 | E: 3 | BS: 16	0.7040	0.6953	0.6996	0.7040	58 min 55 sec
DistilBERT	LR: 2e-05 | E: 2 | BS: 16	0.6919	0.6950	0.6934	0.6919	20 min 56 sec
	LR: 3e-05 | E: 3 | BS: 32	0.6849	0.6874	0.6861	0.6849	29 min 54 sec
	LR: 4e-05 | E: 5 | BS: 32	0.6649	0.6861	0.6753	0.6649	48 min 54 sec
	LR: 2e-05 | E: 3 | BS: 16	0.6962	0.6876	0.6919	0.6962	28 min 23 sec

Varying F1-scores of both models in different hyperparameters shows that the values fall between their respective Precision and Recall values. This indicates that the models were able to maintain a good balance in identifying true positives and making correct predictions. Moreover, values of the metrics suggest that an aggressive approach for both Learning Rates and Epochs resulted in a decrease in model performance. A high learning rate of 4e-05 with a longer 5 epochs led to lower Precision, Recall, and F1-scores compared to parameters with a base rate of 2e-05 on 2 to 3 epochs. This ultimately points to the idea that a slower rate of learning, even with less training runs, allowed the model to learn more effectively than a faster pace of learning.

Table 5. Comparison of BERT and DistilBERT

Hyperparameter			BERT		DistilBERT		BERT vs DistilBERT	
LR	Epoch	Batch Size	Accuracy	Train Time (s)	Accuracy	Train Time (s)	Accuracy	Train Time Ratio
2E-5	2	16	70.84%	2341	69.19%	1256	1.65%	1.86x
3E-5	3	32	70.67%	4296	68.49%	1794	2.17%	2.39x
4E-5	5	32	69.27%	7160	66.49%	2934	2.78%	2.44x
2E-5	3	16	70.40%	3535	69.62%	1703	0.78%	2.07x
							1.84%	**2.19x**

Comparison between BERT and DistilBERT in terms of accuracy and training time in Table 5 presents that with similar hyperparameter settings, DistilBERT trains twice as fast as BERT (2.19x) while maintaining minimal differences in accuracy (1.84%) on average. This makes DistilBERT a viable choice for scenarios where training efficiency is a priority. The result suggests that DistilBERT could achieve comparable accuracy to BERT with less training time in text classification tasks like sentiment analysis. DistilBERT's minimized computational demands due to its reduced parameter size do not sacrifice a huge gap in performance. However, variance in model performance should still be noted to assess performance expectations.

4.4 Model Prediction Results

The BERT model achieving the highest accuracy with the hyperparameter setting (*LR: 2e-05, Epoch: 2, Batch Size: 16*) is applied to predict sentiments on a full data sample for further analysis. The succeeding figures 6 and 7 illustrate the sentiment counts predicted by the trained model for candidates Trump and Harris in each state.

A major takeaway from this distribution is the strong presence of positive sentiments for Kamala Harris in all three states. The prediction for Harris also does not show a sign of extreme negative tone in each state, suggesting that her overall sentiment is balanced and moderate. Comparing this to candidate Trump, he dominated the positive sentiments in Michigan, but this is paired with a high negative result in Arizona. The state of Pennsylvania for both candidates is mostly balanced, but candidate Trump has more divided sentiments. Harris presents more neutral and positive, with significantly lower negative sentiments.

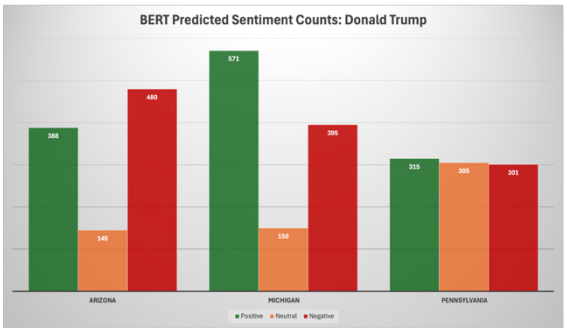

Figure 6. Predicted Sentiment Counts for Donald Trump

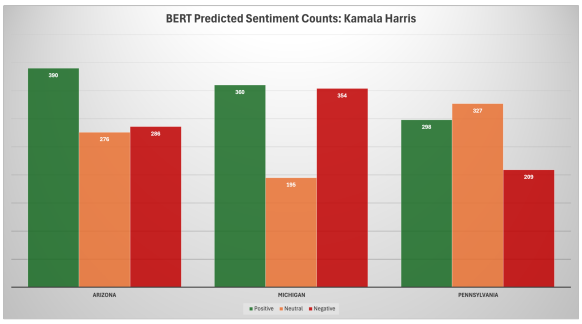

Figure 7. Predicted Sentiment Counts for Kamala Harris

4.5 Word Cloud Analysis

Figure 8 presents both positive and negative world clouds associated with Donald Trump. Positive word cloud highlights mention of "victory", "support", "news", "economy", "win", "policy", and "job". This suggests that discussions framed Trump positively in terms of success, economic strength, and policy impact. Furthermore, it also emphasizes themes of triumph, economic performance, and public backing. On the other hand, the negative sentiment word cloud shows "threat", "job", "day", "court", and "woman" as the prominent keywords, possibly suggesting major concerns related to danger, employment, legal matters, gender issues, and political opposition.

In Figure 9 for candidate Kamala Harris, the positive sentiment word cloud highlights keywords such as "woman", "support", "care", "job", and "policy". These words suggest a strong emphasis on gender representation, backing from supporters, healthcare, employment, and governance. Kamala Harris' depiction of the negative sentiment word cloud includes themes such as "border", "administration", "policy", "interview", and "support". This indicates criticisms on key issues such as immigration policies, her connection to the Biden administration, governance decisions, media interactions, and political backing.

Figure 8. Donald Trump Positive and Negative Word Cloud

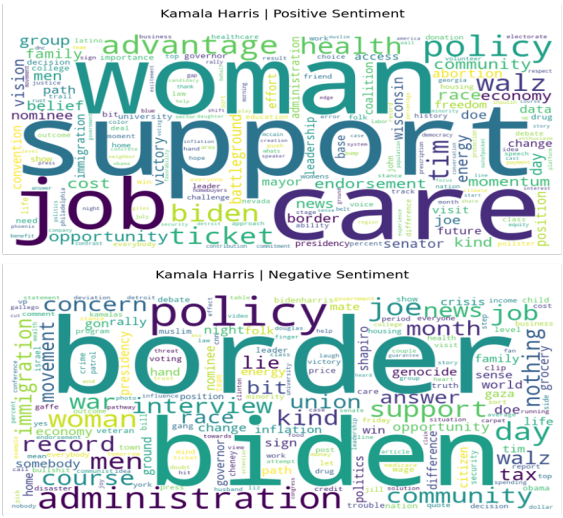

Figure 9. Kamala Harris Positive and Negative Word Cloud

4.6 Actual Vote and Average Sentiment Comparison

To enable a direct comparison between the sentiments analyzed in this study and the actual election outcome, the study first calculated the necessary features to estimate the candidate's proportion of sentiment share in each state. A higher percentage share would suggest a more favorable sentiment toward the candidate. Min-Max normalization is primarily used to rescale the average sentiment values predicted by the model into a normalized scale that allows for a meaningful interpretation. Softmax normalization is also explored to have a probabilistic interpretation that emphasizes sentiment differences and influence. Based on the calculated

features, the results indicated no significant differences in terms of the overall sentiment share for each candidate, regardless of the normalization method used. Significantly, Democratic candidate Kamala Harris has demonstrated higher sentiment proportion shares compared to Republican candidate Donald Trump in states like Arizona and Pennsylvania.

Table 6. Min-Max Calculated Features

State	Presidential Candidate	Average Sentiments	Min-Max Normalized	Candidate's Proportion of Sentiment Share
Arizona	Donald Trump	-0.0908	0.4545	45.04%
	Kamala Harris	0.1092	0.5546	54.96%
Michigan	Donald Trump	0.1577	0.5788	53.49%
	Kamala Harris	0.0066	0.5033	46.51%
Pennsylvania	Donald Trump	0.0152	0.5076	47.84%
	Kamala Harris	0.1067	0.5533	52.16%

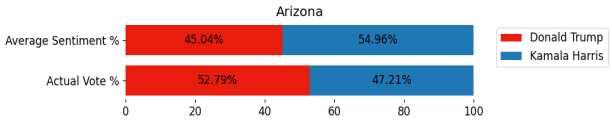

Figure 10. Average Sentiment Share in Arizona vs Actual Vote

The average sentiment percentage in the state of Arizona presents Kamala Harris leading with an estimated 54.96% sentiment shares, while Donald Trump yielded 45.04%, respectively. However, the actual vote results reveal that Trump won Arizona with 52.79% of the actual vote, compared to Harris' 47.21%. This discrepancy indicates that positive sentiment towards Harris in Arizona did not correspond with the actual votes.

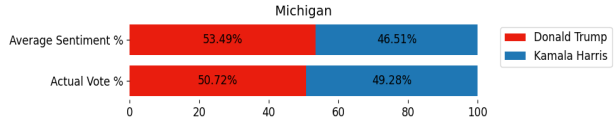

Figure 11. Average Sentiment Share in Michigan vs Actual Vote

In the state of Michigan, both the average sentiment shares and the actual vote results aligned, with Trump holding the majority in both percentages. Trump yielded a higher 53.49% average sentiment share in contrast to Harris' 46.51%. Michigan is the only state where the sentiment analysis result correctly reflected the election outcome.

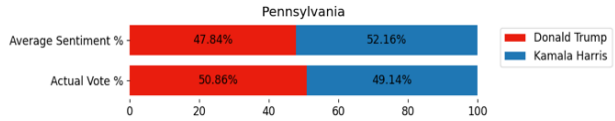

Figure 12. Average Sentiment Share in Pennsylvania vs Actual Vote

In Pennsylvania, Kamala Harris received a higher average sentiment share with 52.16% compared to Trump's 47.84%. However, Trump secured more actual votes recorded at 50.86% versus Harris' 49.14%. Similar to Arizona, this shows a mismatch between sentiment trends and voting behavior.

5. CONCLUSION AND FUTURE WORK

Sentiment Analysis is essential in understanding public opinion, particularly in the political domain. Following the 2024 U.S. presidential election, this study utilized the model BERT to analyze the sentiments of spoken content in video data discussing Donald Trump and Kamala Harris in three swing states (Arizona, Michigan, and Pennsylvania). The videos were transcribed and underwent an in-depth text preprocessing phase. Its main objectives include performance assessment of BERT, model comparison with DistilBERT, and validating the analyzed sentiments against the actual election outcomes at a state level.

The findings revealed that BERT achieved the highest sentiment classification accuracy of 70.84%, with DistilBERT reaching up to 69.62%. In the results presented, it is observed that while BERT models attained higher accuracies than DistilBERT models in all examined hyperparameters, DistilBERT still trains twice as fast as BERT while maintaining minimal difference in their performance. This emphasizes that in cases where efficiency is prioritized, DistilBERT is deemed the better choice. Furthermore, the study also found that despite the sentiments providing relevant insights about voters' opinions, it does not fully reflect the actual election outcomes.

In the future, the researchers suggest exploring methods to enhance data quality, such as refining text preprocessing, and sentence filtering that would also identify entity mentions of specific states. Additionally, the researchers also recommend enhancing the model's capability in classifying neutral sentiments where it involves interpretations of ambiguous language and context.

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